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Toward the Era of Agentic Embodied Intelligence

A2-MEAS: Learnable Correction Codes Link LLMs to Control

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In one sentence: strong action model + learnable correction codes set a new public-leaderboard record on long-horizon chores

First, build a strong action model, A2-WLA (VLA + motion-aware history + implicit world model). When the robot errs, the MetaBrain LLM and diagnostic agents issue corrections backed by spatial evidence, **encoded as a "correction code" injected into the frozen action expert**, with zero parameter updates at deployment. BEHAVIOR 2026 (100 tasks): Q 0.6411 / SR 45.40%.

0.6411

BEHAVIOR, 100 tasks

Full success rate 45.40%; 88/100 tasks succeed at least once

95.05%

RoboTwin 2.0 avg.

A2-WLA alone 93.98% (π0.5: 79.75%)

57.92%

Real robot, 12-task macro avg.

π0.5: 26.67%; high-level correction off

14→94%

Coffee task: post-failure recovery

Step 2; Step 7: 8%→82% (100 trials per condition)

4.03×

Training throughput

427.1 vs 106 samples/s, same 8-GPU hardware

Note: on 100 tasks, "single→multi-model" adds +0.144 Q, while "multi-model→full system (agentic correction)" adds +0.037 Q; the agent layer's gain on the official hidden set is small.

Problem: knowing to "retry" is not knowing "what to do differently next time"

1

Long horizon = failures inevitable

BEHAVIOR demos avg. 5.86 min, eval episodes 8.40 min; one partial success can make later steps infeasible

2

High-level replanning only sets "what"

Language skill selection and feedback replanning can decide to "retry", but not how approach direction, alignment or gripper timing should change

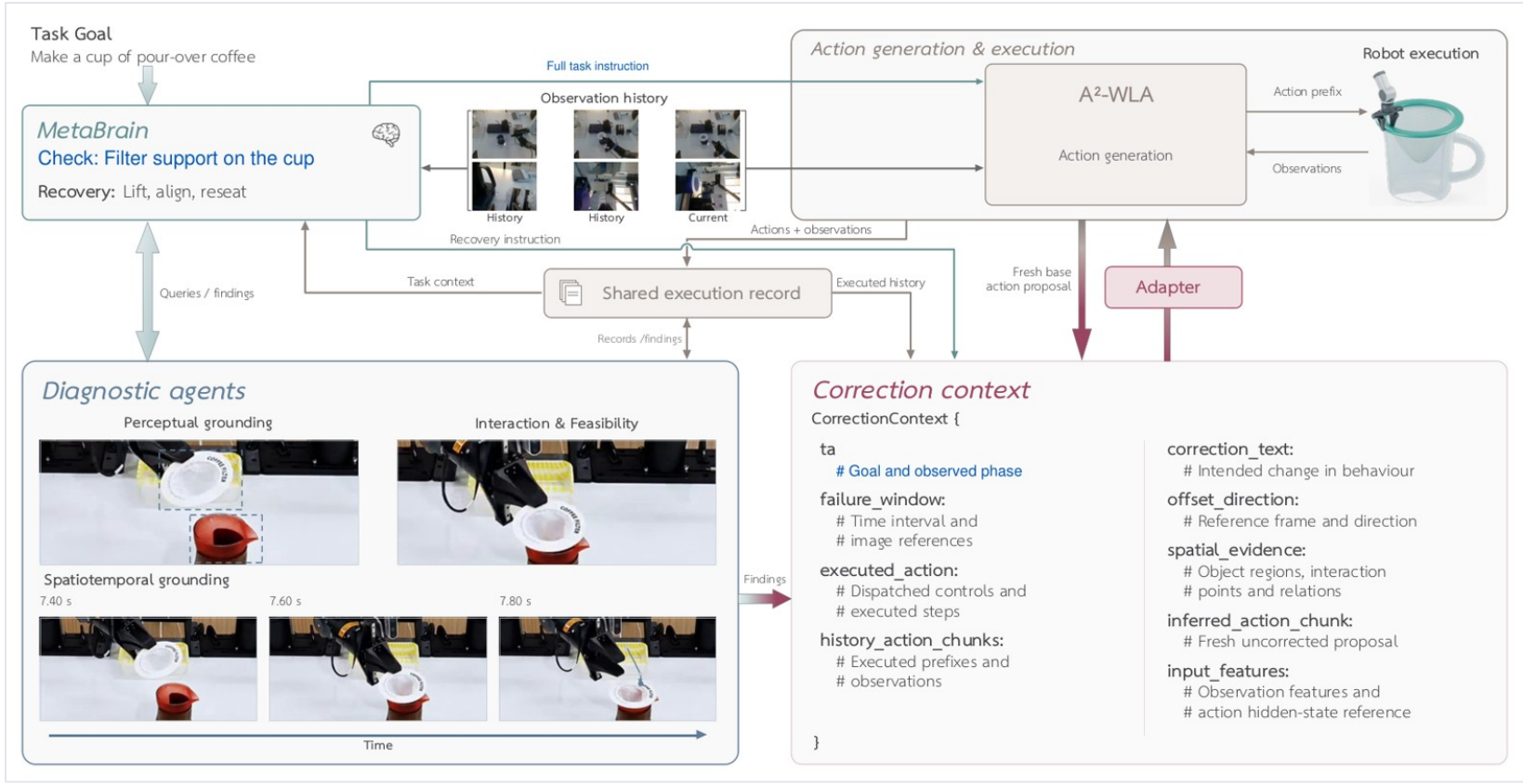
3

A learned interface is needed

Feed correction intent + spatial evidence + execution history into action generation so diagnosis actually changes the next trajectory

Closed-loop recovery of a failed filter placement: slip → lift → align → place & verify (Fig. 2)

A2-MEAS: three agent groups share one execution record to jointly control one robot



MetaBrain

Qwen3.8-27B Coordinator

Track goals • verify progress • diagnose failures • write recovery instructions (with reference frame & direction)

Diagnostic Agents

Percep./spatiotemp./interaction

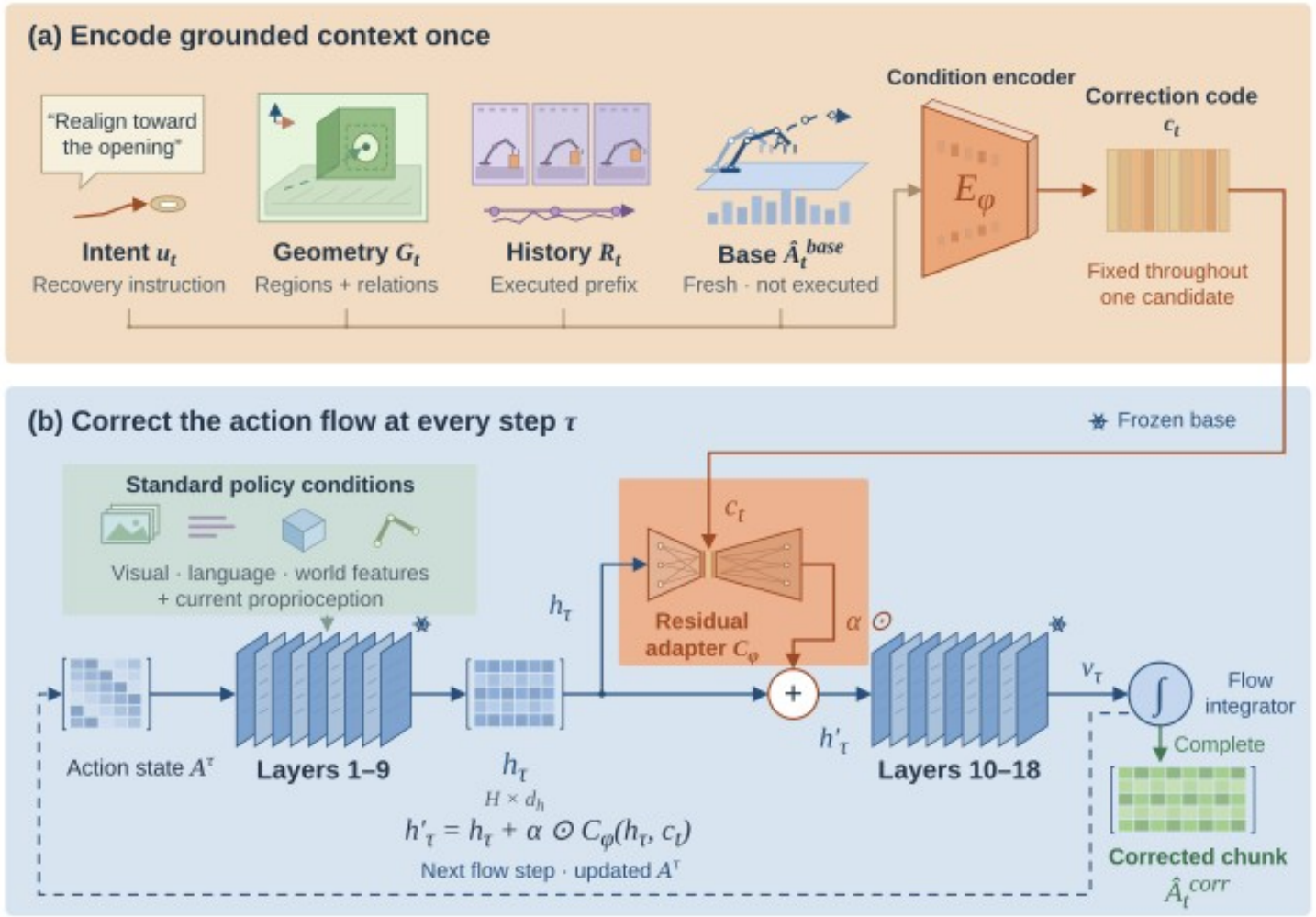
DINO, SAM, tracking, depth, robot state; 6 query types; separate "negative" from "unknown"

A2-WLA

2B VLM + World Model + Action Expert

Generates continuous action chunks & local progress; correction branch takes the correction code

Core: correction code c_t injects residuals inside the action expert, zero parameter updates at deployment



Four inputs → one continuous code

$$c_t = E_\phi(u_t, G_t, R_t, \hat{A}_t^{base})$$

u_t correction language • G_t spatial evidence • R_t executed history • Â^base current unexecuted proposal

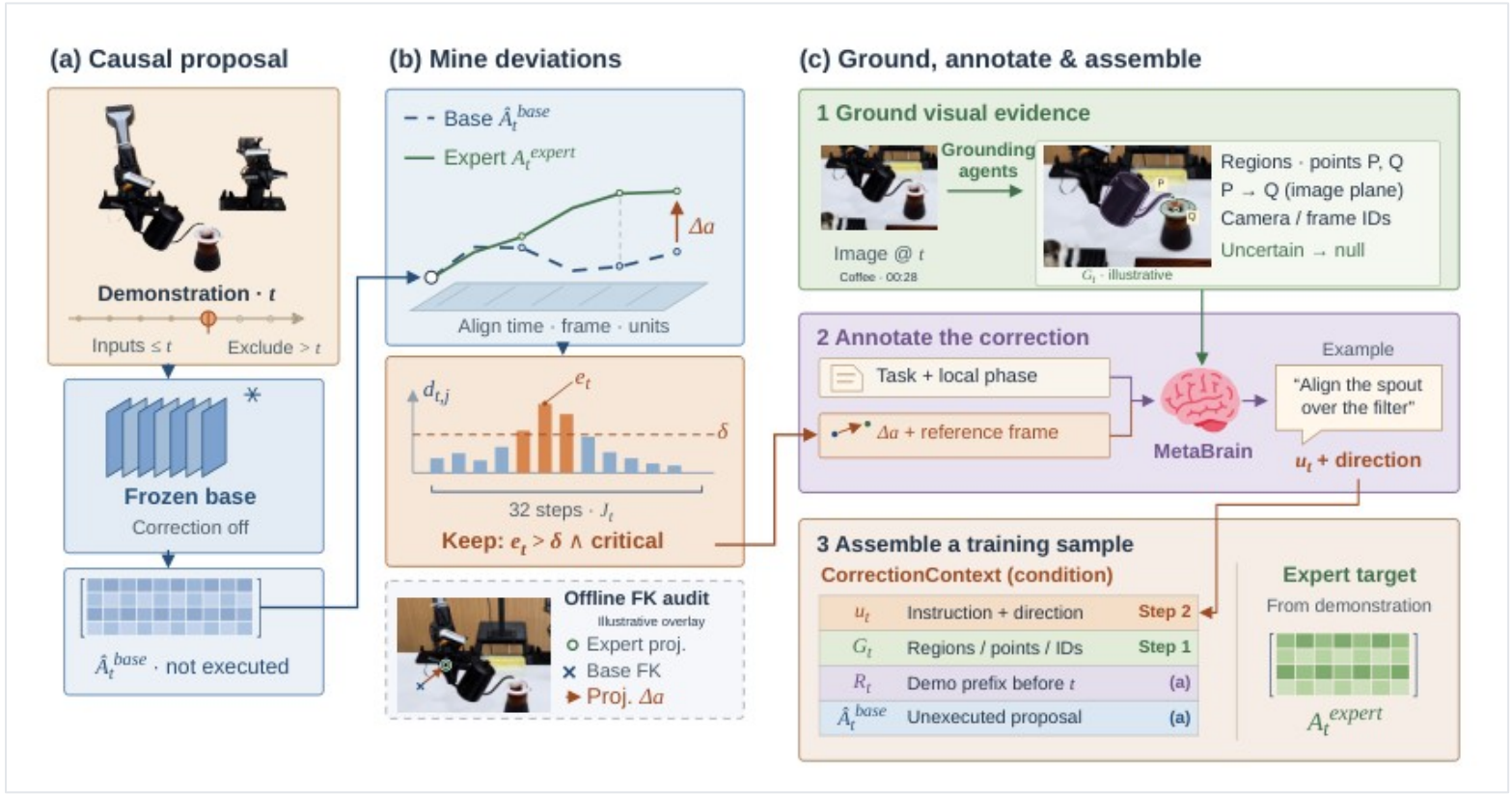
The same instruction yields different corrections under different evidence/history/proposals

Residuals injected after layers 9 / 18

$$h'_\tau = h_\tau + \alpha \odot C_\phi(h_\tau, c_t)$$

- Recomputed from the current hidden state at every flow step → position-dependent fixes within a chunk
- Zero-initialized output projection, α=0.1 → initially equivalent to the original policy
- Bottleneck 256; correction branch < 100M params
- Base A2-WLA frozen; all parameters fixed at deployment

Correction supervision comes from offline-mined "policy vs. expert" deviations, not real failures



① Sample proposals at key moments

At skill switches / pre-contact approach / gripper boundaries, frozen policy generates unexecuted chunks

② Mine large deviations

Align 32-step windows with expert; keep windows with e_t > δ at key steps

③ Grounding + annotation

Open-vocab grounding + SAM; MetaBrain writes corrections from deviation direction

④ Train only correction branch ϕ

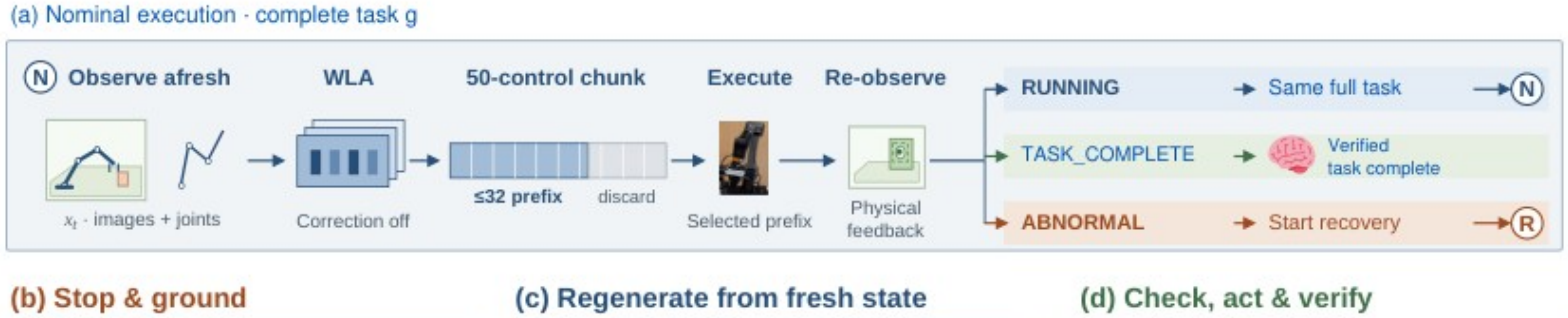
Target = expert continuation; flow-matching loss; θ frozen

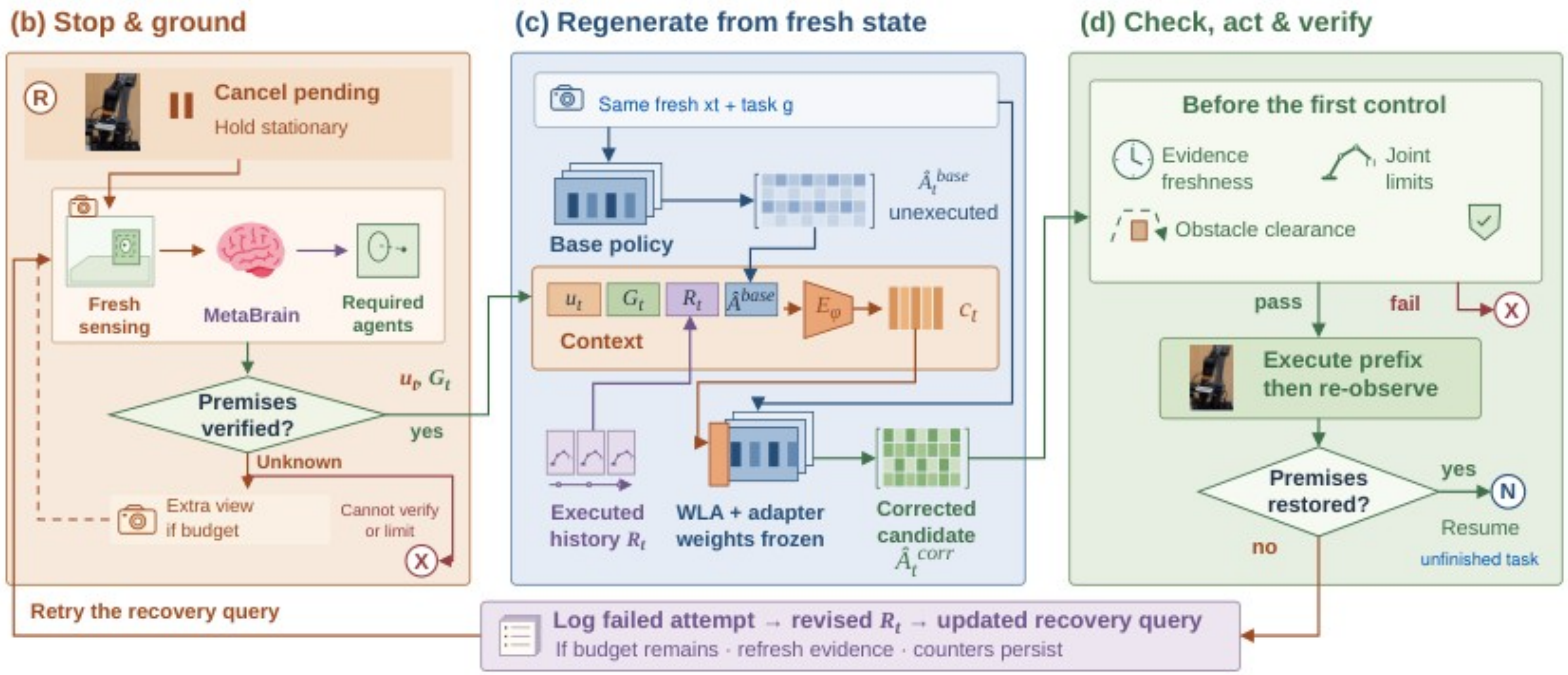
⩽ 2% corpus: 39 h

Limit: deviation ≠ real failure; distribution gap

Source: Fig. 5, Eq. (4), Sec. 2.5

Closed-loop recovery: pause → verify preconditions → regenerate → safety check → execute & verify





Budget per anomaly

2
Extra views
5 s each

3
Physical retries
Count not reset

10 s
MetaBrain timeout
Per request

60 s
Total recovery cap
Exit if exceeded

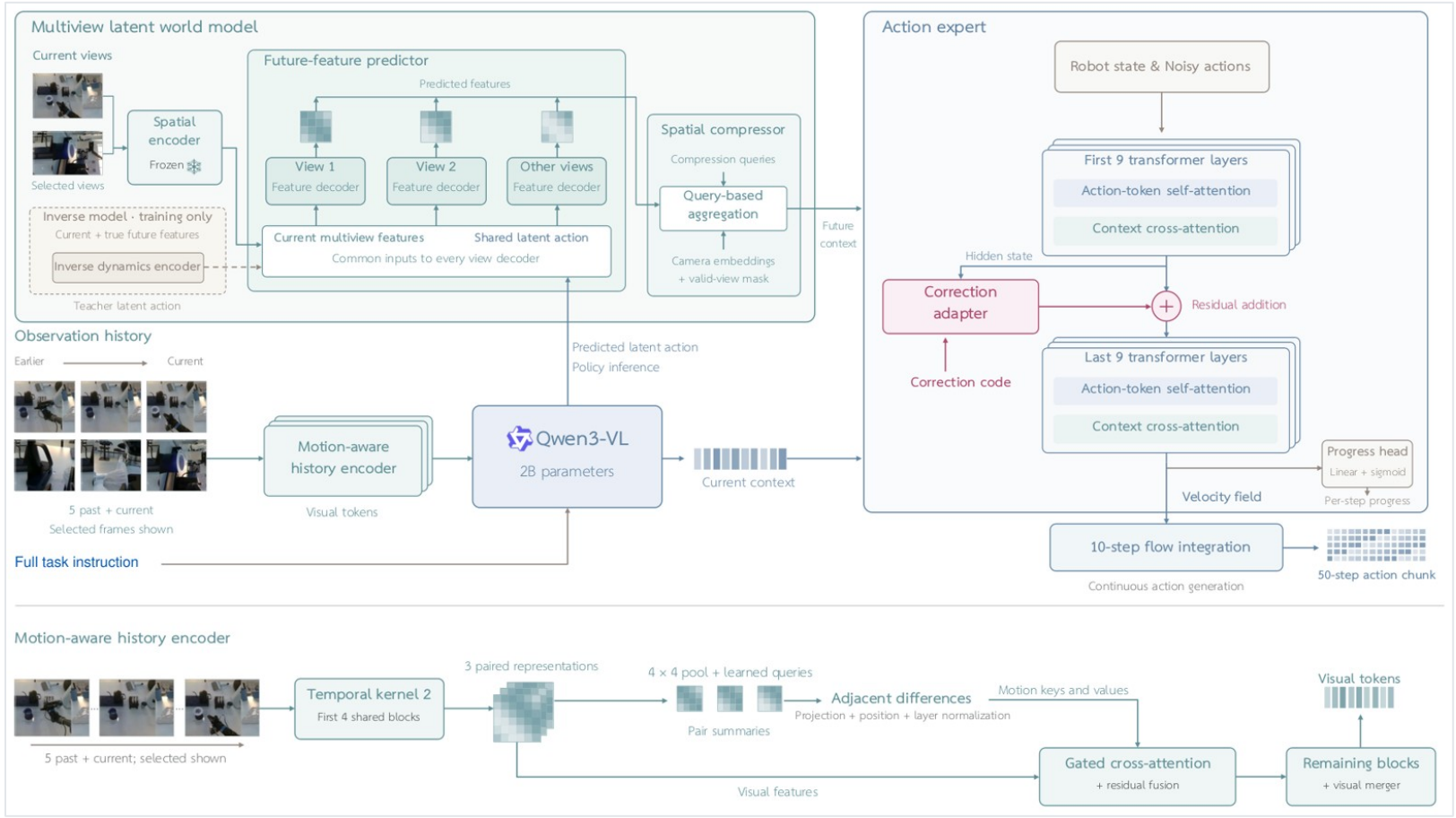
Negative ≠ Unknown

Obstacle seen (negative) → reroute

Occluded (unknown) → add a view

Predict 50 steps, execute only first ⩽32, then re-observe

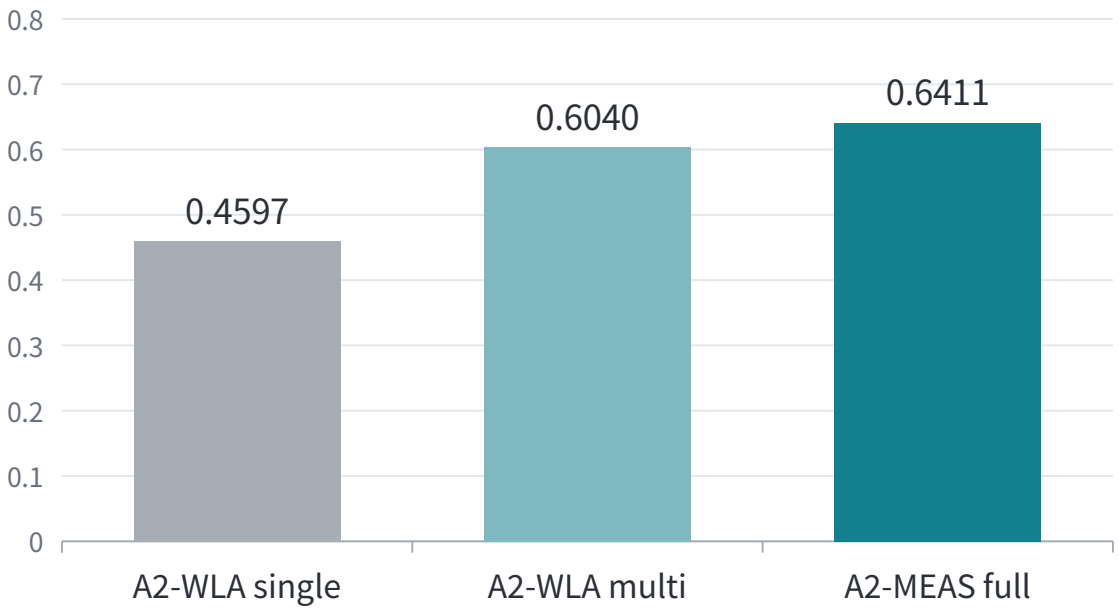
A2-WLA: VLA + motion-aware history + multi-view implicit world model



Backbone	Qwen3-VL 2B
History	5 history frames + current per view; adjacent diffs as motion K/V
World model	Inverse dynamics yields shared latent action z^*; at deployment VLM predicts $\hat{z} \rightarrow$ future context C^+
Action expert	~300M, 18 layers; even layers attend C, odd C^+; 10 flow steps → 50-step chunk
Training	① World model ② VLM (FAST / multi-candidate / latent action / VQA) ③ joint flow matching
Data tricks	Skill boundaries ±60 frames ×5 sampling; proprio dropout 0.3→0.05

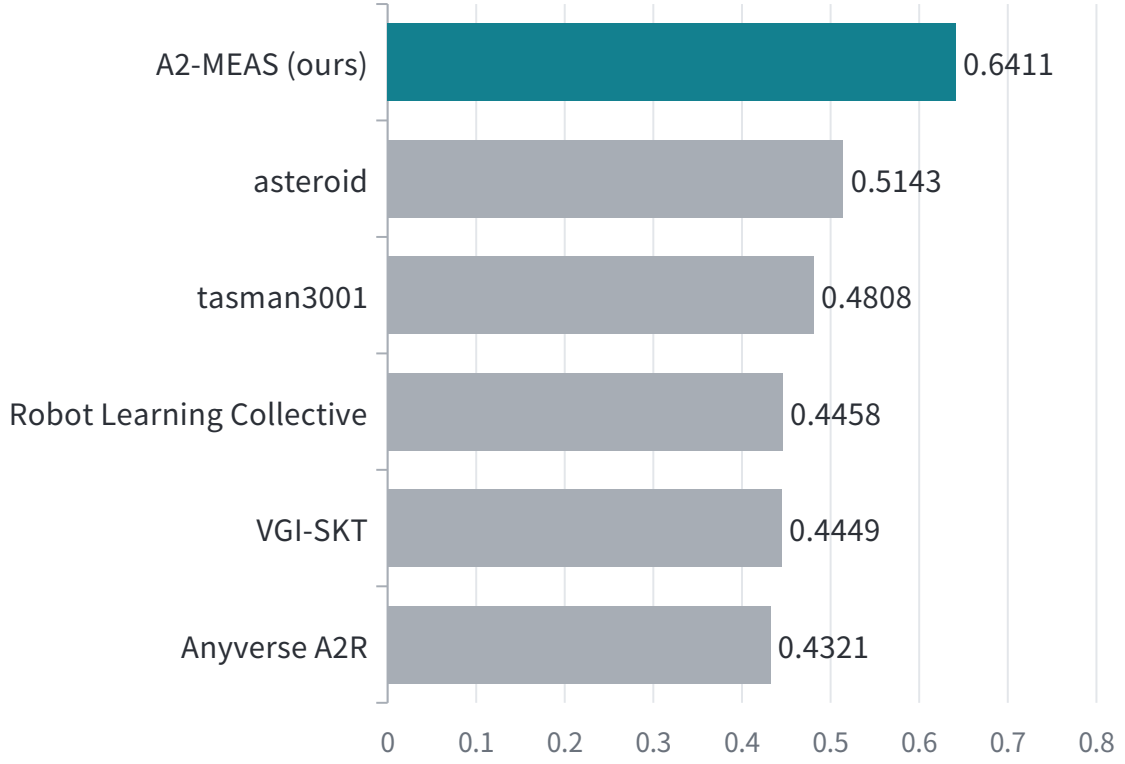
BEHAVIOR 2026 public set: Q 0.6411 / SR 45.40%, well ahead of the self-reported leaderboard

Three configs (100 tasks × 10 instances)



	A2-WLA single	A2-WLA multi	A2-MEAS full
Success rate	27.20%	40.60%	**45.40%**
Tasks covered	70	80	**88**

Public-set self-reported Q-score (excerpt)



Official hidden set (24 GB limit, coordinator swapped to Qwen3.5-2B): A2-MEAS 0.4975 / 32.05% vs A2-WLA alone 0.4951 / 30.95%; the agent layer's gain is small.

Source: Tables 5, 6 (self-reported leaderboard, accessed 2026-10-31)

Gain breakdown: mostly from the action model and per-task model selection; agentic correction adds +0.037

0.4597

A2-WLA single

One checkpoint runs all 100 tasks

+0.1443

Per-task model (0.6040)

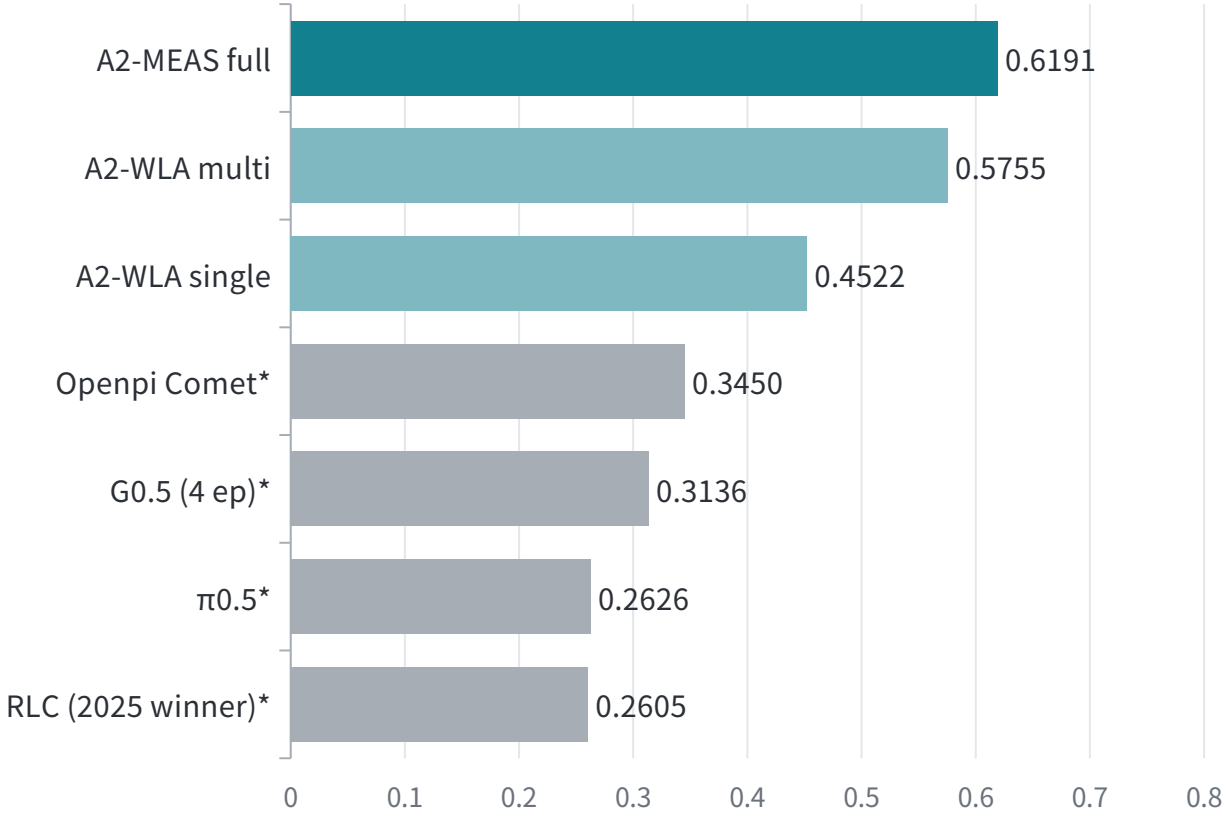
Task-specific policy library; mapping set by prior evals

+0.0371

Agentic correction (0.6411)

MetaBrain + diagnosis + correction-code Adapter

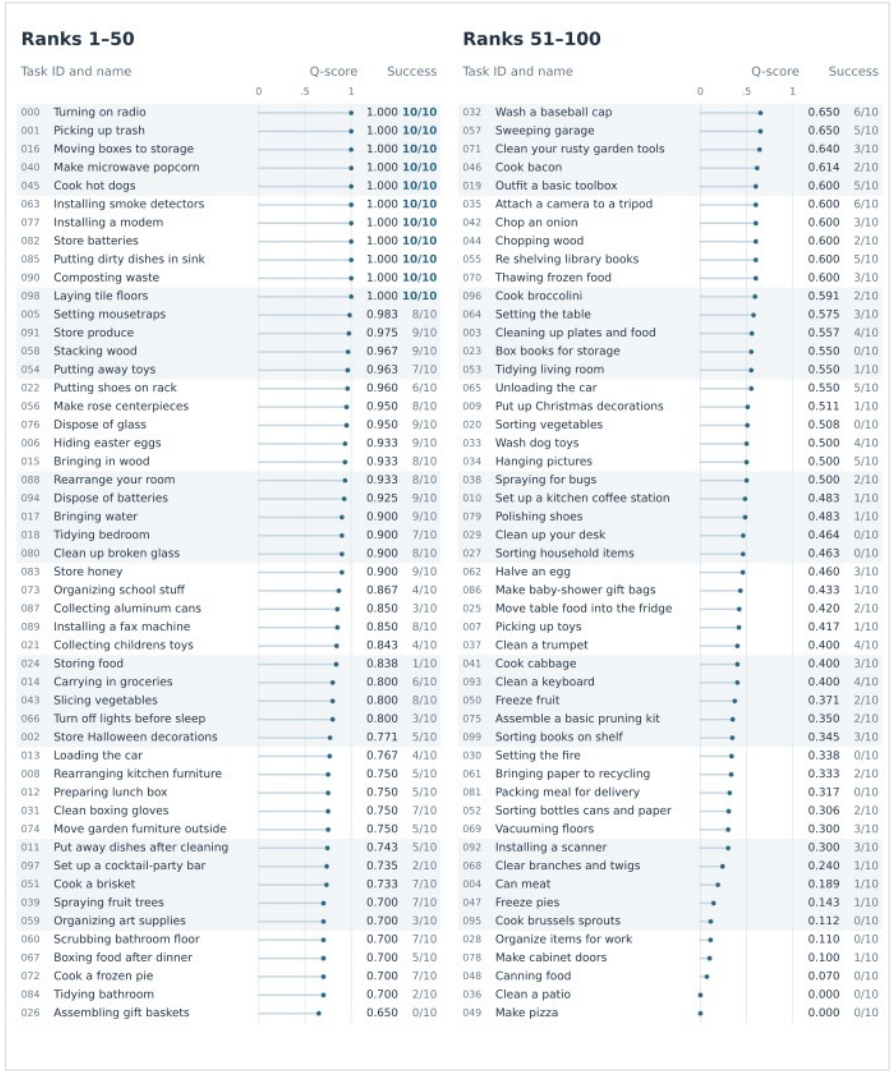
50-task subset Q-score vs BEHAVIOR 2025 public results



* BEHAVIOR 2025 public-set results: different benchmark from the 2026 subset, different training setup; "2.21×" (0.5755/0.2605) is only a rough reference.

Source: Tables 5, 8 (50-task subset & 2025 public results)

Per-task distribution: broad coverage, concentrated weak spots; "partial" ≠ "complete"



Ranks 1–50

Task ID and name	Q-score	Success
000 Turning on radio	1.000	10/10
001 Picking up trash	1.000	10/10
016 Moving boxes to storage	1.000	10/10
040 Make microwave popcorn	1.000	10/10
045 Cook hot dogs	1.000	10/10
063 Installing smoke detectors	1.000	10/10
077 Installing a modem	1.000	10/10
082 Store batteries	1.000	10/10
085 Putting dirty dishes in sink	1.000	10/10
090 Composting waste	1.000	10/10
098 Laying tile floors	1.000	10/10
005 Setting mousetraps	0.983	8/10
091 Store produce	0.975	9/10
058 Stacking wood	0.967	9/10
054 Putting away toys	0.963	7/10
022 Putting shoes on rack	0.960	6/10
056 Make rose centerpieces	0.950	8/10
076 Dispose of glass	0.950	9/10
006 Hiding easter eggs	0.933	9/10
015 Bringing in wood	0.933	8/10
088 Rearrange your room	0.933	8/10
094 Dispose of batteries	0.925	9/10
017 Bringing water	0.900	9/10
018 Tidying bedroom	0.900	7/10
080 Clean up broken glass	0.900	8/10
083 Store honey	0.900	9/10
073 Organizing school stuff	0.867	4/10
087 Collecting aluminum cans	0.850	3/10
089 Installing a fax machine	0.850	8/10
021 Collecting childrens toys	0.843	4/10
024 Storing food	0.838	1/10
014 Carrying in groceries	0.800	6/10
043 Slicing vegetables	0.800	8/10
066 Turn off lights before sleep	0.800	3/10
002 Store Halloween decorations	0.771	5/10
013 Loading the car	0.767	4/10
008 Rearranging kitchen furniture	0.750	5/10
012 Preparing lunch box	0.750	5/10
031 Clean boxing gloves	0.750	7/10
074 Move garden furniture outside	0.750	5/10
011 Put away dishes after cleaning	0.743	5/10
097 Set up a cocktail-party bar	0.735	2/10
051 Cook a brisket	0.733	7/10
039 Spraying fruit trees	0.700	7/10
059 Organizing art supplies	0.700	3/10
060 Scrubbing bathroom floor	0.700	7/10
067 Boxing food after dinner	0.700	5/10
072 Cook a frozen pie	0.700	7/10
084 Tidying bathroom	0.700	2/10
026 Assembling gift baskets	0.650	0/10

Ranks 51–100

Task ID and name	Q-score	Success
032 Wash a baseball cap	0.650	6/10
057 Sweeping garage	0.650	5/10
071 Clean your rusty garden tools	0.640	3/10
046 Cook bacon	0.614	2/10
019 Outfit a basic toolbox	0.600	5/10
035 Attach a camera to a tripod	0.600	6/10
042 Chop an onion	0.600	3/10
044 Chopping wood	0.600	2/10
055 Re shelving library books	0.600	5/10
070 Thawing frozen food	0.600	3/10
096 Cook broccolini	0.591	2/10
064 Setting the table	0.575	3/10
003 Cleaning up plates and food	0.557	4/10
023 Box books for storage	0.550	0/10
053 Tidying living room	0.550	1/10
065 Unloading the car	0.550	5/10
009 Put up Christmas decorations	0.511	1/10
020 Sorting vegetables	0.508	0/10
033 Wash dog toys	0.500	4/10
034 Hanging pictures	0.500	5/10
038 Spraying for bugs	0.500	2/10
010 Set up a kitchen coffee station	0.483	1/10
079 Polishing shoes	0.483	1/10
029 Clean up your desk	0.464	0/10
027 Sorting household items	0.463	0/10
062 Halve an egg	0.460	3/10
086 Make baby-shower gift bags	0.433	1/10
025 Move table food into the fridge	0.420	2/10
007 Picking up toys	0.417	1/10
037 Clean a trumpet	0.400	4/10
041 Cook cabbage	0.400	3/10
093 Clean a keyboard	0.400	4/10
050 Freeze fruit	0.371	2/10
075 Assemble a basic pruning kit	0.350	2/10
099 Sorting books on shelf	0.345	3/10
030 Setting the fire	0.338	0/10
061 Bringing paper to recycling	0.333	2/10
081 Packing meal for delivery	0.317	0/10
052 Sorting bottles cans and paper	0.306	2/10
069 Vacuuming floors	0.300	3/10
092 Installing a scanner	0.300	3/10
068 Clear branches and twigs	0.240	1/10
004 Can meat	0.189	1/10
047 Freeze pies	0.143	1/10
095 Cook brussels sprouts	0.112	0/10
028 Organize items for work	0.110	0/10
078 Make cabinet doors	0.100	1/10
048 Canning food	0.070	0/10
036 Clean a patio	0.000	0/10
049 Make pizza	0.000	0/10

11

tasks at 10/10 success

Radio on, hot dogs, modem setup, tiling, etc.

8

tasks with Q < 0.2

Pizza-making, patio cleaning: goal satisfaction 0

5 / 34

Q ⩾ 0.8 yet SR ⩽ 40%

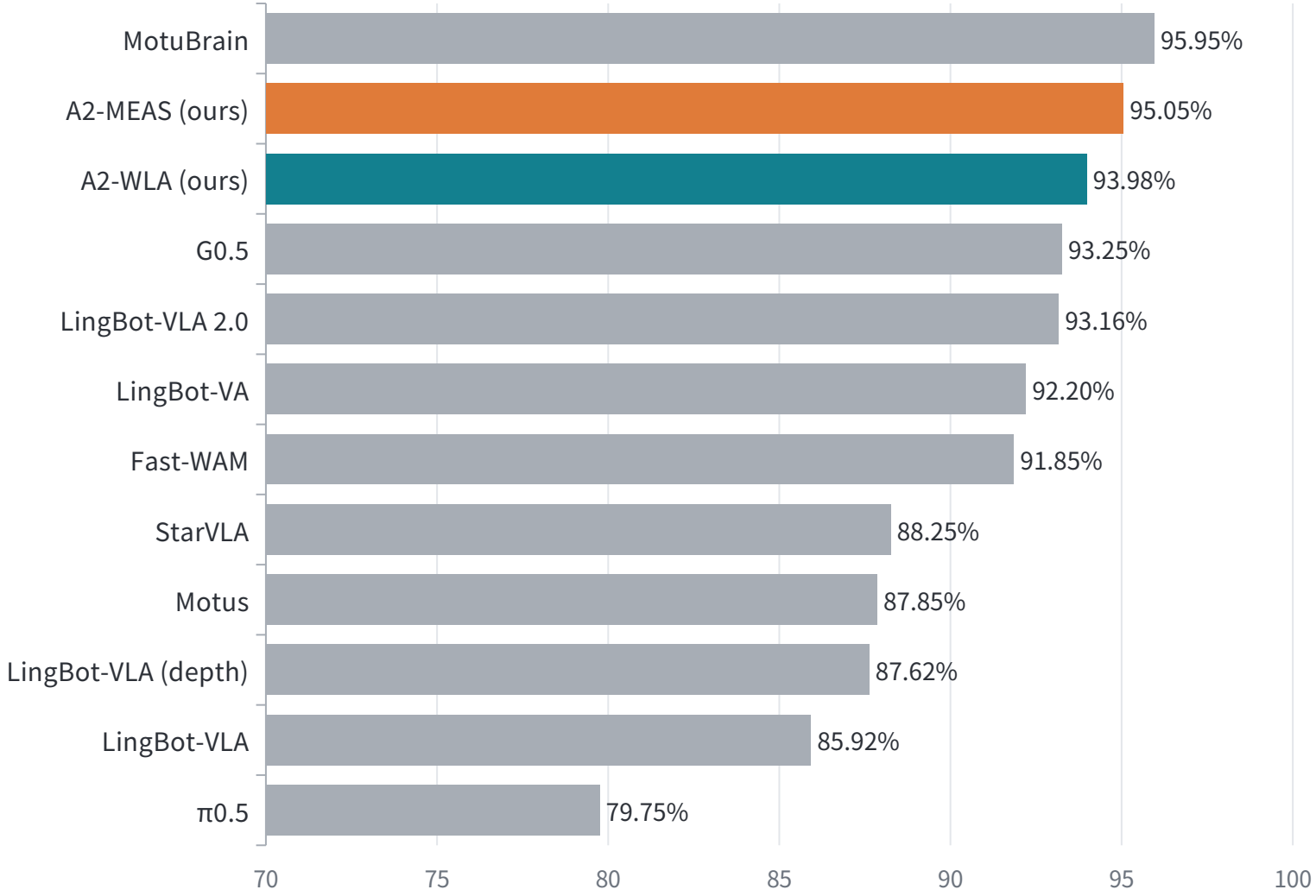
Store food: Q 0.8375, only 1/10 success

10 / 12

Zero-success tasks still score Q > 0

Gift basket 0.65, pack books 0.55

RoboTwin 2.0: A2-WLA 93.98%, full system 95.05%; benchmark near saturation



+14.23 pp

A2-WLA vs π0.5

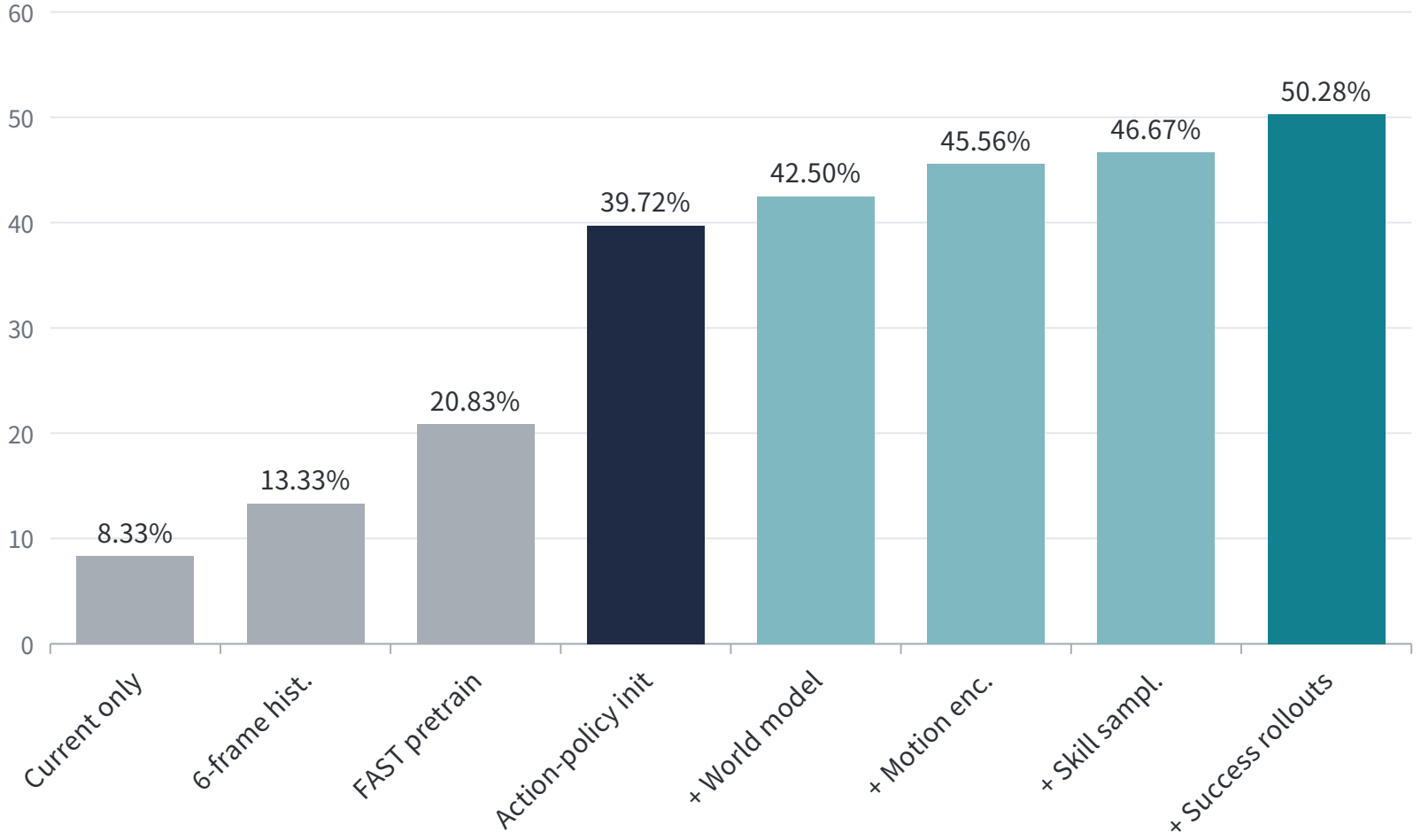
+1.07 pp

Gain from agentic correction

- Clean 94.26% / Randomized 93.70% （A2-WLA）
- Below MotuBrain 95.95% (different protocol)
- SR near 95%; limited discrimination

Source: Table 7 (RoboTwin 2.0, 50 bimanual tasks)

Base-model ablation: action pretraining matters most; world model / motion history / data add up



Gray: obs./init. comparison Navy: action-pretraining baseline Teal: added components (SR)

+18.89 pp

Action-policy init vs FAST

20.83% → 39.72%

0.4846→0.5905

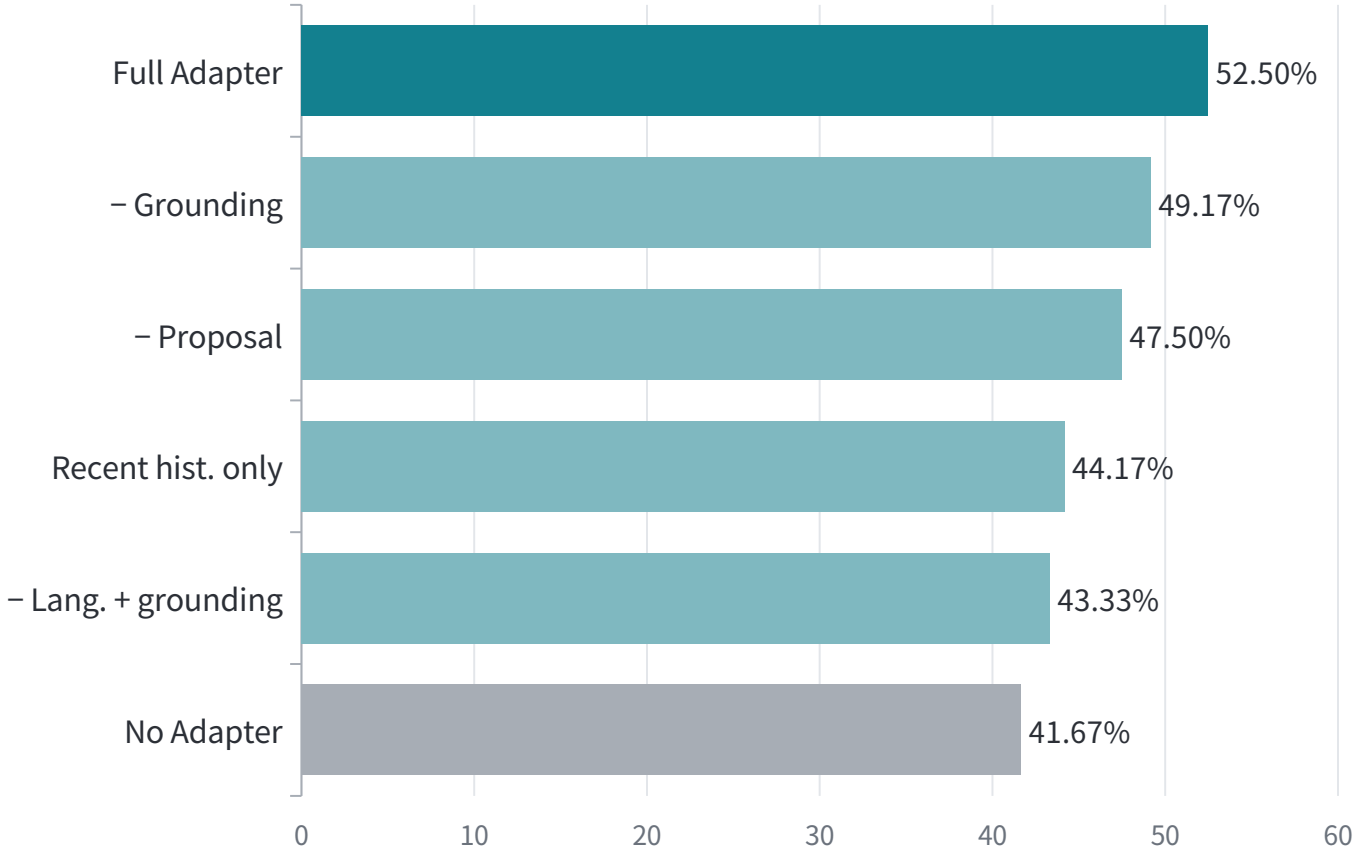
Q-score with added components

SR 39.72% → 50.28%

- World model: Q +0.0337
- Motion-aware encoding: +0.0096 more
- Skill-boundary sampling: +0.0267 more
- 2,400 successful rollouts: +0.0359 more

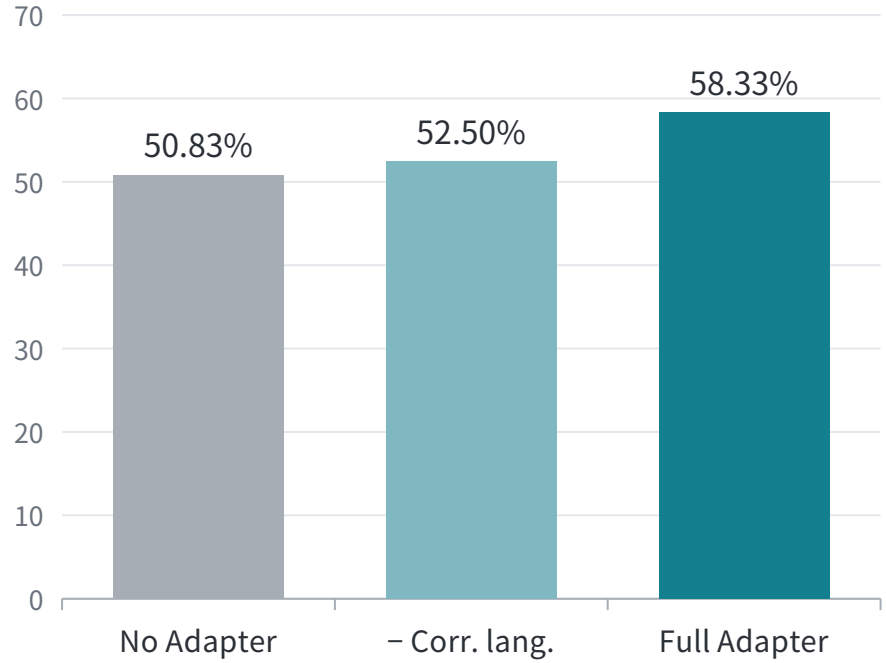
Every correction-code input helps; still +7.5 pp on disjoint instances

Conditioning ablation (first 10 instances) • SR



Q-score: no Adapter 0.5310 → full 0.6108 (+0.0798)

Disjoint instances (last 10) • SR

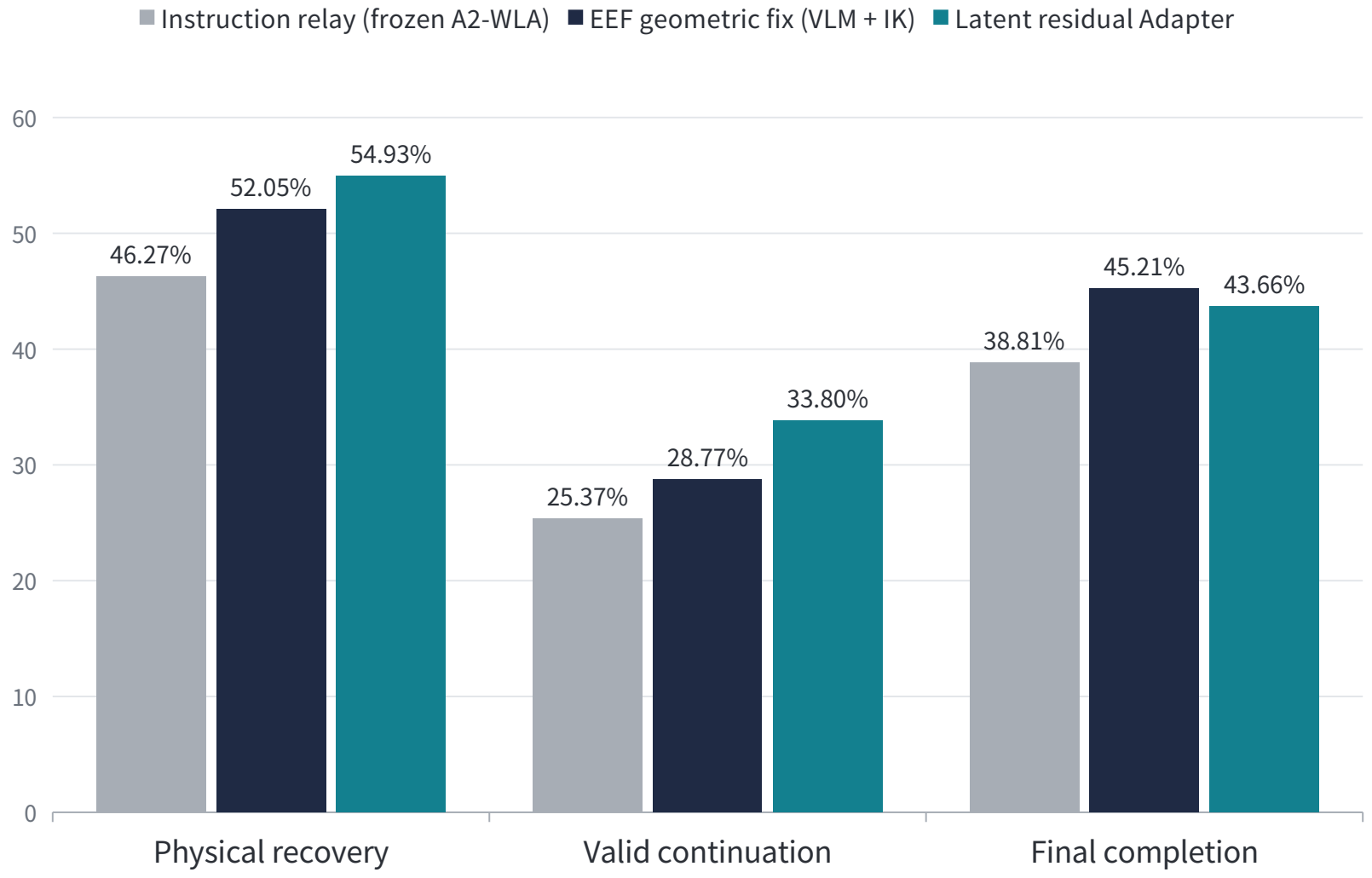


+7.50 pp **61 → 70 successful episodes**

Q: 0.6185 → 0.6782; 0.6336 w/o correction language

Source: Table 12, Fig. 14 (12-task correction subset, 120 episodes each)

Latent Adapter has the highest physical recovery, but lower final completion than VLM + IK



Q-score (all 120 episodes)

Instruction relay	0.5668
VLM + IK	**0.6083**
Adapter	**0.6108**

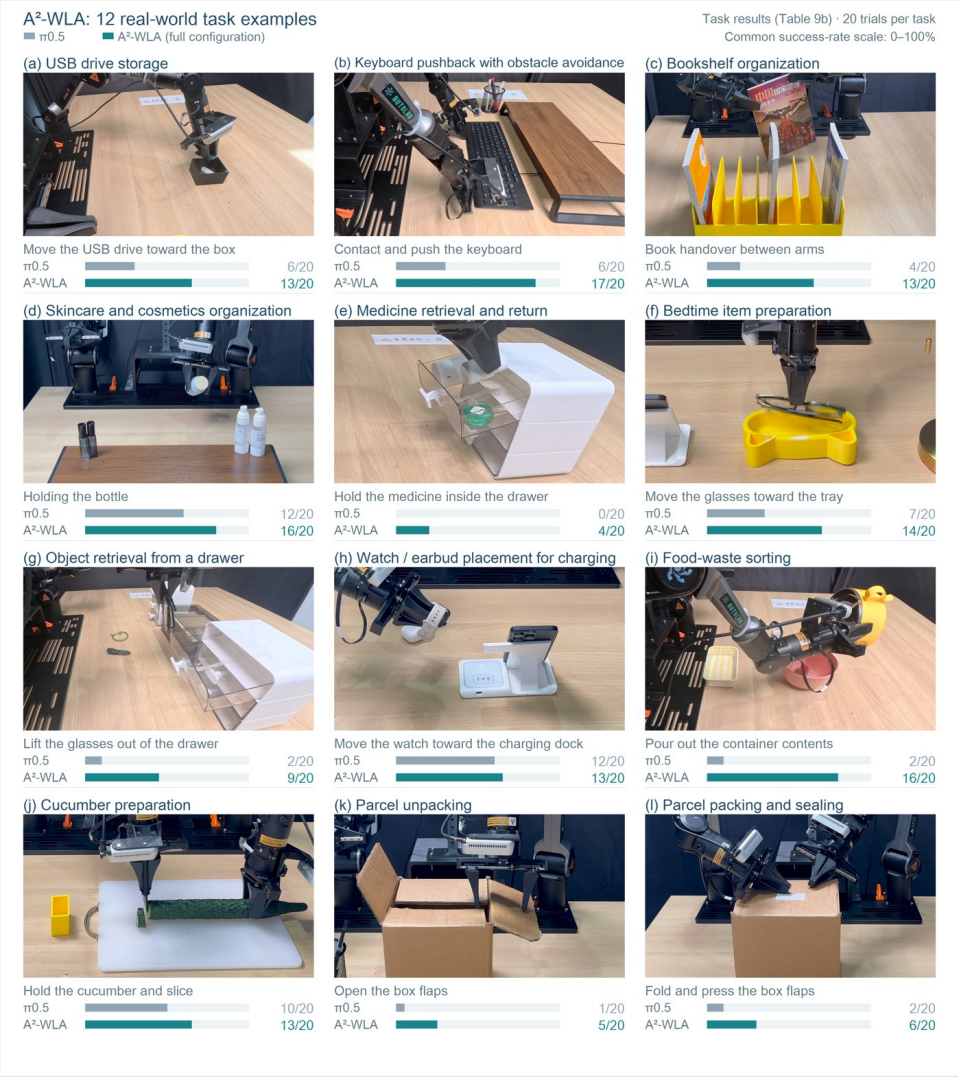
Takeaways

- Better local recovery ≠ more completed tasks
- Interfaces trigger on different episode sets; unpaired comparison
- Params / compute not matched

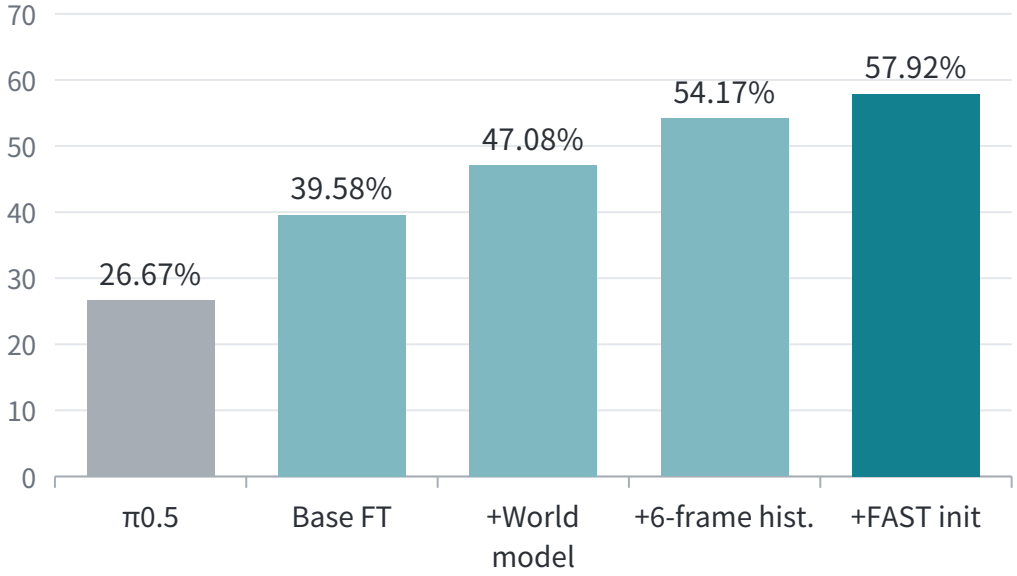
→ Weak evidence that "latent beats geometric correction"

Source: Table 13 (12-task correction subset)

Real robot, 12 tasks: A2-WLA 57.92% vs π0.5 26.67%, ahead on all 12



Macro-avg. SR (12 tasks)

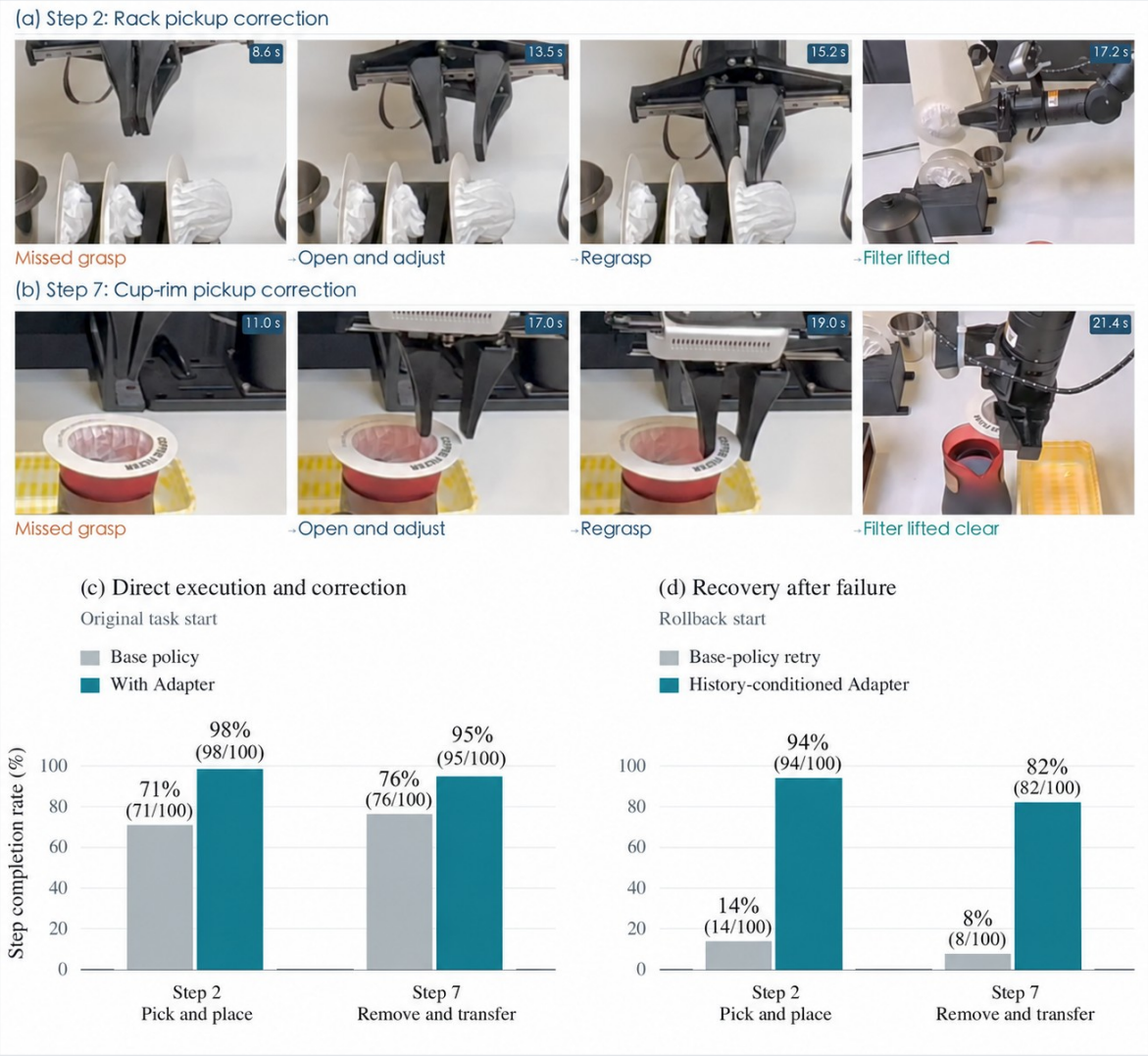


Largest gains / still hard

Food-waste sorting	2 → 16	Pick & place meds	0 → 4
Kbd. push (obst.)	6 → 17	Package unboxing	1 → 5
Bookshelf tidying	4 → 13	Pack & seal box	2 → 6

Source: Fig. 12, Table 9 (20 trials per task per config, high-level correction off)

Pour-over coffee: recovery after failure rollback 14% → 94%, 8% → 82%



After failure & rollback

Step 2 • Place filter on cup rim

14% → **94%**

Step 7 • Remove & transfer

8% → **82%**

Base-policy retry → history-conditioned Adapter

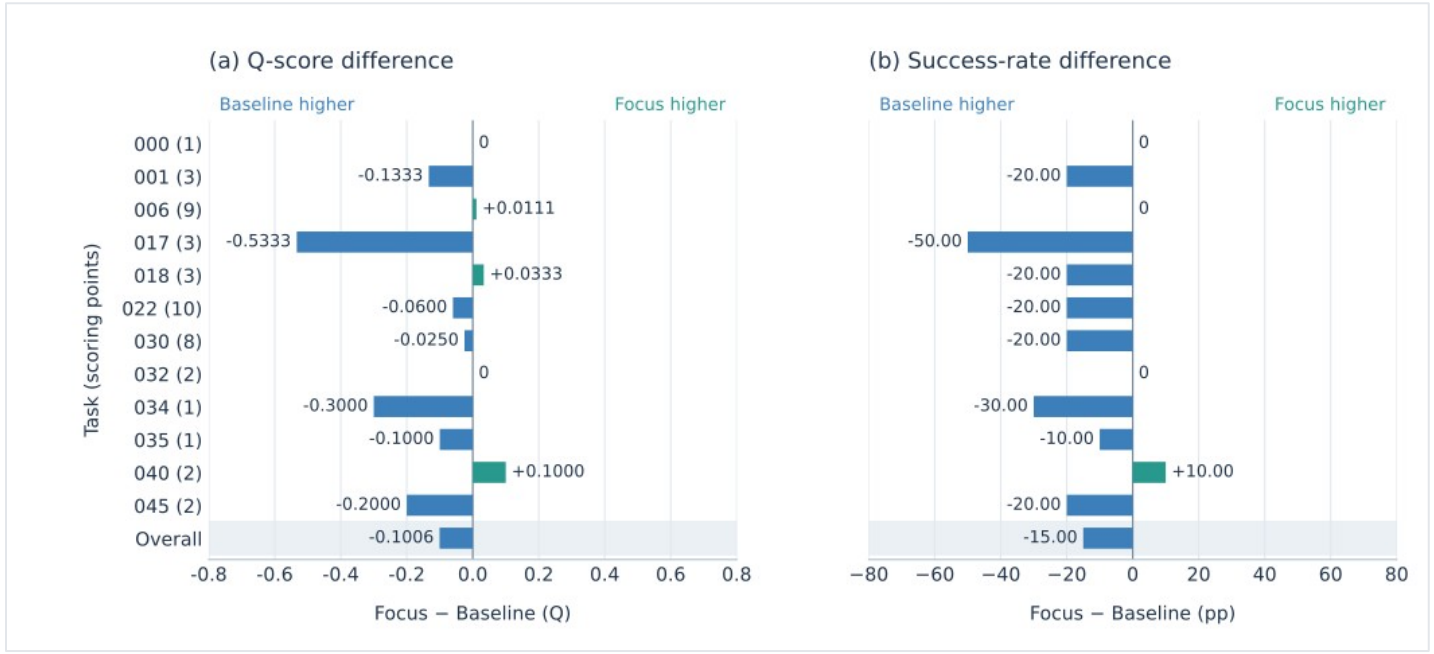
Direct execution from original start: 71→98%, 76→95%

Note: real robot uses a simplified Adapter

Conditioned only on observations + ⩽64-step failure history; ~860K params; residual added before the output head; full four-input correction code not validated.

Source: Fig. 13, Sec. 5.4.2 (100 trials per condition)

Subtask scheduling performs worse: long-horizon skill should be learned into the policy itself



Per-task difference, Focus − full-task conditioning (left: Q; right: SR in pp) (Fig. 16)

−0.1006

Focus: Q change

−15 pp

Focus: SR change

Four schedules, fixed model (Q / SR)

F-VLM online focus	**0.4411 / 30.83%**
F-Global full task only	**0.4201 / 30.00%**
F-Linear fixed order	**0.4115 / 25.00%**
F-Adaptive reorderable	**0.3785 / 27.50%**

Why? Sparse transition context

75/100 tasks have procedural variants; of 6,849 transition combos conditioned on the previous two steps, 62.8% have ⩽5 supporting demos.

Limits of recovery: base policy recovers on its own but times out; physical damage may be irreversible

Spontaneous recovery without Adapter (Fig. 18)

Glove drops

Bends to pick up

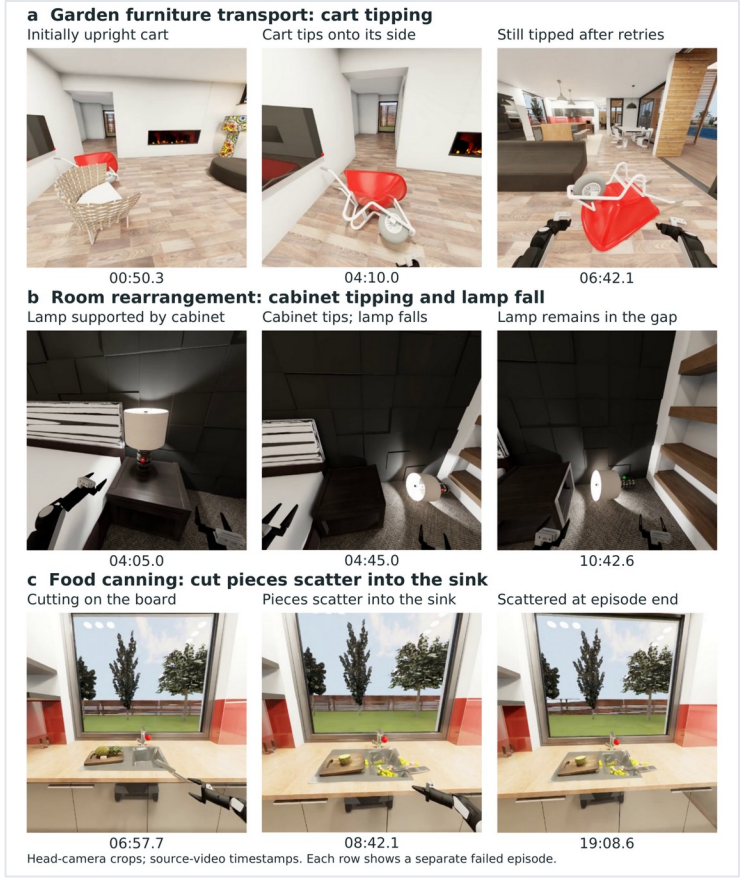
Into washer ✓

Regrasp inverted bottle (timeout)

Regrasp corn (timeout)

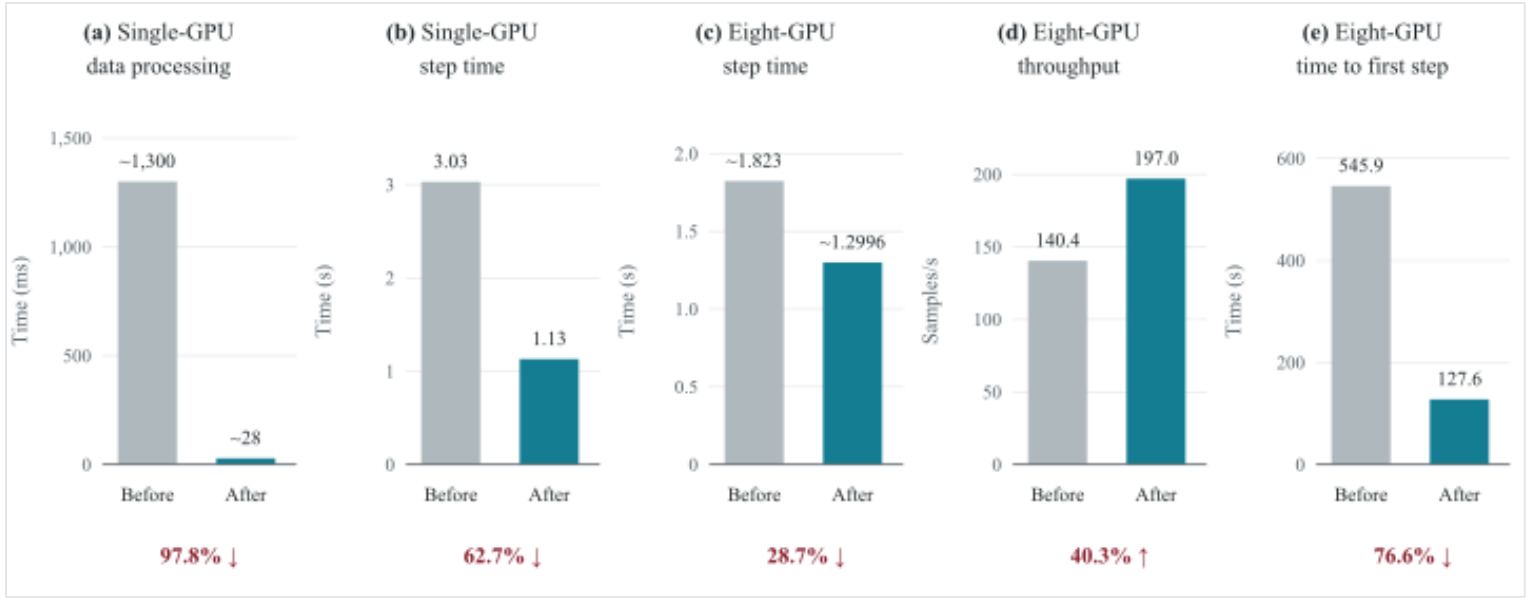
→ Recoverable ≠ efficiently recovered: targeted correction guidance is needed

Irreversible physical state changes (Fig. 19)



Scene damaged → recovery may exceed reach / joint / gripper limits

Infrastructure: 4.03× training throughput; full 1,000-episode eval in 18.1 hours



Data proc. ↓97.8%, 1-GPU step ↓62.7%, 8-GPU tput ↑40.3%, 1st step ↓76.6% (Fig. 8)

427.1 /s

8-GPU training throughput (4.03× π0.5)

π0.5 ref. 106/s; own batch sizes

18.1 h

1,000-episode eval (8 GPUs × 3 workers)

≈55.2 episodes/h; 7.7× sim real-time

~98%

64-GPU scaling efficiency

8.8× eval vs ideal 1,275 h / 8

Source: Fig. 8, Table 4, Sec. 4

Verdict: solid engineering and strong numbers, but "agentic" gains need independent verification

✓ Strengths

Unified correction interface: heterogeneous agents → CorrectionContext → correction code into action expert; zero updates at deployment

Simple action model is strong: pretraining + history + world model + data sampling; consistently ahead in sim and real

Measurable correction gains: +10.83 pp with fixed policy, +7.50 pp on disjoint instances

Complete engineering: 4.03× training, full eval within a day; code and models promised open-source

Candid negative results: Focus scheduling worse, VLM+IK higher completion, small hidden-set gain

? Concerns

Gain source: multi-model +0.144 Q ≫ agents +0.037; task→model mapping set by prior evals

Cross-year comparison: "2.21× the 2025 winner" is not the same benchmark

Official hidden set: agents add only +0.0024 Q with 2B coordinator

Real robot: only the simplified history-conditioned Adapter validated

Interface comparison: unpaired; final completion below VLM+IK

Method limits: correction requires pausing; trained on deviations, not real failures; some experiments single-run, no variance

Three Takeaways

1

Learn capability into the policy first

A strong action model (pretraining + history + world model) drives most long-horizon performance

2

Agents handle the critical points

Verify + correct; codes let heterogeneous reasoning steer control with no deploy updates

3

Whole-system view & independent verification

Policy, orchestration, control and hardware co-evolve; agent-layer gains await replication after open-sourcing

arXiv 2611.09124 • Code (incremental open-source promised): github.com/AutoLab-SAI-SJTU/A2MEAS